

Linear Algebra

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To the Instructor

Linear algebra is a topic that can be taught at many different levels, depending upon the sophistication of the audience. These notes were initially developed for a one semester sophomore-junior level linear algebra course for a group of students who had a familiarity with vectors from multi-variable calculus and physics. But these students had had no prior course in vector algebra per se, and had minimal experience proving theorems. They completed Chapter 1 and most of Chapters 2 and 3.

Subsequently I taught a one semester senior-beginning graduate level linear algebra course. Most of these students had experience proving theorems, and had had an elementary course in matrix algebra and vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$. These students were able to begin with and complete Chapter 2, and do Chapter 3 and a new Chapter 4 as well. As a result they went through a solid axiomatic development of finite-dimensional linear spaces, inner product spaces, and linear transformations.

In Chapter 1 students draw on a basic experience with three-dimensional vectors to begin thinking of them as forming a linear space. Here I avoid formalism and try to build a working intuition about vectors from both an algebraic and a geometric point of view: norm as length, addition and scalar multiplication, linear combinations, span and bases. As applications of these ideas we look at vector descriptions of lines and solutions to systems of linear equations. I find it important to frequently repeat the italicized principle at the top of page two, as this is what distinguishes this course for most other mathematics courses they have had:

In this course you will learn about linear algebra by solving a carefully designed sequence of problems. Unlike other mathematics courses you have had in the past, solving a problem in this course will always have two components.

1. *Find a solution.*
2. *Explain how you know that your solution is correct.*

This will help prepare you to use mathematics in the future, when there will not be someone at hand to tell you if your solution is correct. In those situations, it will be up to you to determine whether or not it is correct.

For most students I find that Problems 1 and 2 bring this principle home quickly. They have learned the formula for Problem 2, and want to apply it to both.

The second chapter provides the basic structure of finite-dimensional linear spaces, introducing enough examples to illustrate the topics that arise. I use a slight variation of the usual axioms so that, for example, $1\mathbf{P} = \mathbf{P}$ is Theorem 22 rather than an axiom. I advise

you to think carefully about the definition of “span”. In these notes I have first defined the span of a finite set, let the students work with this notion a bit, and then extended the definition to all sets. But I have found, with some classes who are struggling with the concept of an abstract linear space, the extended definition becomes too abstract. For those classes I recommend omitting the extended definition and modifying or omitting the few subsequent problems that depend upon it. Almost all of what is done here concerns only finite bases and finite-dimensional linear spaces, and therefore does not depend upon the extended definition of “span”.

Problem 30 can lead to interesting discussions since they don’t yet have the tools to solve it, but they are certainly in a position to think about it and make a reasonable conjecture, particularly if they have done Problem 29. Problem 36 is a step in the right direction, and it is usually profitable to discuss whether or not Problem 30 is solved by solving Problem 36. Problem 30 arises again as Problem 48, where now they appreciate the solution as an easy application of the Basis Theorem. For experiences like this it is important to pass out the text as it is needed, rather than in a single packet. After they prove Lemma 48 and Theorem 40, I like to raise the issue as to whether this would lead to a proof that every linear space (as, for example, $\mathbb{C}[0, 1]$) has a basis. My intention is to raise this question but not press it unless some student(s) take a real interest in it. The chapter ends with applications of Gaussian elimination, which was introduced without a name in Chapter 1.

A central theme of the third chapter is that statements which are true in $\mathbb{R}^2$ and $\mathbb{R}^3$ and can be expressed in the language of inner products are generally true in all inner product spaces. This theme is a nice illustration of the process of mathematical abstraction. The main effort of the chapter is to prove the Gram-Schmidt Theorem, which is built up in many steps. Early in the chapter we prove the Cauchy-Schwartz Inequality using a simple trick that appears as a highly un-intuitive rabbit-out-of-a-hat, and then use it to prove the Triangle Inequality. As an application of the Gram-Schmidt Theorem we get very straightforward proofs of these two results from our deeper understanding of the structure of these spaces.

The fourth chapter introduces the notion of isomorphic spaces through the observation that all two-dimensional inner product spaces ‘look just like’ $\mathbb{R}^2$. This leads to the notion of a linear transformation, which I believe should be defined by implications and then proven (Lemma 74) to be equivalent to equations. We conclude by seeing how linear transformations of finite-dimensional spaces are represented by matrices.

I have taught the elementary version of this course (Ch 1,2,3) to close to 30 students and the advanced version (Ch 2,3,4) to as few as 10, both at SUNY New Paltz. I give two exams, a midterm and a final, each counting 25% of the grade. Class presentations and student portfolios each count another 25%.

Class time is primarily organized around class presentations. I present definitions and statements of problems and theorems in an interactive discussion with the class. I try to motivate the content, raise questions, and connect ideas. The students are then left to solve the problems and prove the theorems on their own, outside of class, without consulting other sources. At the start of class students mark on a sheet which items they are ready to present. I choose students to present who have presented the least. For classes with over 20 enrolled, I normally have several students put up work simultaneously. Then I go through each proof/solution individually with the class. Students make good mistakes that are generally instructive to their classmates. A student presenting an incorrect proof has the opportunity to correct it for the next class meeting.

All students are required to keep a portfolio consisting of a final correct solution to each problem and proof of each theorem. For the portfolio work, they can consult each other or me, and they can make multiple attempts to get it right. For weaker students the process of writing up a proof/solution from their own class notes can be as much of a discovery experience as any. But they are aided by the fact that they have already seen the work done in class.

Portfolios are required to be readable, complete and correct. Accordingly, the final portfolio grade is either 25% or 0%. I have them turned in periodically so that I can record their progress. In doing so I choose sample items to read carefully; the number of them being determined by the time I have to do so. Occasionally I omit a selected item from class presentation and have all students do it themselves for the portfolio.

I offer these notes as a starting point for the instructor who would like to teach an inquiry-based course in linear algebra. These notes have evolved over many iterations and now work well for most of the mathematics students at SUNY New Paltz, a moderately competitive 4 year state college. Depending on your audience, you may well need to modify them, either adding more material and more difficult problems or skipping some of the material and filling in some easier problems. Regardless of your audience, you may choose to replace some parts of this guide with topics, examples or problems that match your personal interests.

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Chapter 1

Vectors in 3-Space

A **vector** is an ordered triple $\mathbf{P} = (a_1, a_2, a_3)$ where a_1, a_2 and a_3 are in the set $\mathbb{R}$ of real numbers. The special vector $\mathbf{0} := (0, 0, 0)$ is called the **origin**. The vector $\mathbf{P}$ is illustrated by either

- the point in space with coordinates $x = a_1, y = a_2$ and $z = a_3$,
- the arrow drawn from the origin $\mathbf{0} = (0, 0, 0)$ to the point $\mathbf{P} = (a_1, a_2, a_3)$ or
- any arrow with the same length and direction as this arrow, that is, drawn from any point (x_1, x_2, x_3) to the point $(a_1 + x_1, a_2 + x_2, a_3 + x_3)$.

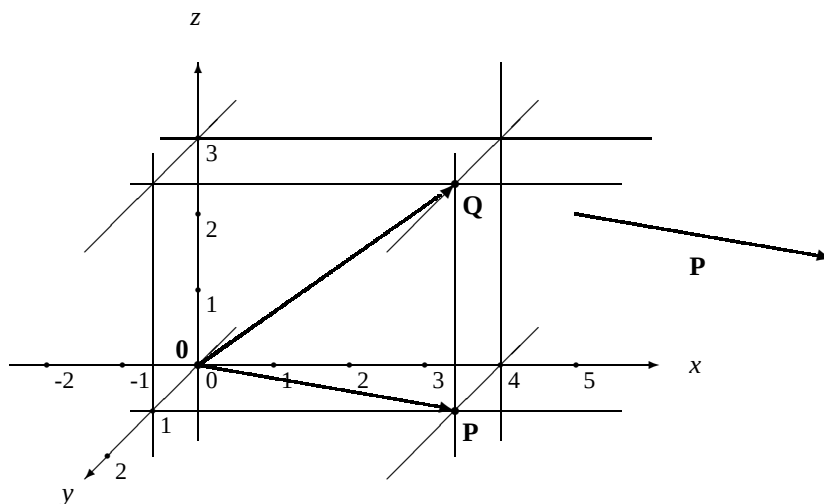


Figure 1.1: $\mathbf{P} = (4, 1, 0)$, $\mathbf{Q} = (4, 1, 3)$

Vectors are typically used to represent physical quantities such as location in space, velocity, acceleration, force, momentum or torque. Thought of as an arrow, a vector $\mathbf{P}$ has both

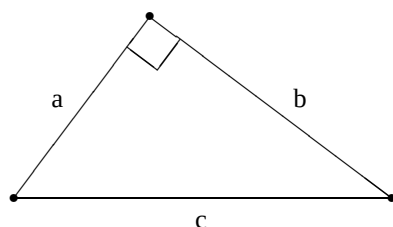
a direction and a length. If $\mathbf{P}$ represents one of these physical quantities, its direction tells us how it is oriented and its length gives us its magnitude. The length or magnitude of $\mathbf{P}$ is called the **norm** of $\mathbf{P}$ and is denoted by $\|\mathbf{P}\|$.

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1. Find a solution.
2. Explain how you know that your solution is correct.

This will help prepare you to use mathematics in the future, when there will not be someone at hand to tell you if your solution is correct. In those situations, it will be up to you to determine whether or not it is correct.

Your first task will be to find a way to compute the norm of a vector using a well known theorem from geometry.



Pythagorean Theorem

$$a^2 + b^2 = c^2$$

In the following two problems, carefully draw and label the relevant right triangles to show how you are using this theorem.

Problem 1. Use Figure 1.1 to find $\|\mathbf{P}\|$ and then use $\|\mathbf{P}\|$ to find $\|\mathbf{Q}\|$.

Problem 2. Now let $\mathbf{P} := (a_1, a_2, 0)$ and $\mathbf{Q} := (a_1, a_2, a_3)$ where a_1, a_2 and a_3 are arbitrary numbers. Find a formula for $\|\mathbf{P}\|$ and use it to find a general formula for $\|\mathbf{Q}\|$:

If $\mathbf{Q} := (a_1, a_2, a_3)$, then $\|\mathbf{Q}\| =$.

Problem 3. Referring again to the vectors $\mathbf{P}$ and $\mathbf{Q}$ in Figure 1.1, find the vectors $\mathbf{U}$ and $\mathbf{V}$ that are in the direction of $\mathbf{Q}$ such that $\|\mathbf{U}\| = 3\|\mathbf{Q}\|$ and $\|\mathbf{V}\| = 12$. Use your formula from Problem 2 to check your answers.

Given vectors (thought of as arrows) $\mathbf{P} = (a_1, a_2, a_3)$, $\mathbf{Q} = (b_1, b_2, b_3)$ and a number t , we define

- the **sum** $\mathbf{P} + \mathbf{Q} = (a_1, a_2, a_3) + (b_1, b_2, b_3) := (a_1 + b_1, a_2 + b_2, a_3 + b_3)$,
- the **scalar product** $t\mathbf{P} = t(a_1, a_2, a_3) = (ta_1, ta_2, ta_3)$.

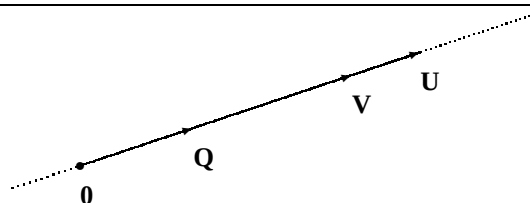


Figure 1.2: Vectors in the direction of $\mathbf{Q}$.

Problem 4. Each of these algebraic concepts has a nice geometric interpretation. To see this, let $\mathbf{P} = (a_1, a_2, 0)$ and $\mathbf{Q} = (b_1, b_2, 0)$. Draw an xy -plane (the points with z -coordinate 0), and indicate in it the locations of points $\mathbf{P}$, $\mathbf{Q}$, $(-1)\mathbf{P}$, $3\mathbf{Q}$ and $\mathbf{P} + \mathbf{Q}$.

For vectors $\mathbf{P}$ and $\mathbf{Q}$, the **difference** $\mathbf{Q} - \mathbf{P}$ is the unique vector that, when added to $\mathbf{P}$, gives us $\mathbf{Q}$. We can describe the difference both geometrically and algebraically, each in two different ways.

Problem 5. Draw an origin $\mathbf{0}$ and then draw two vectors (as arrows) $\mathbf{P}$ and $\mathbf{Q}$ emanating from it.

- (i) Draw an arrow that represents the difference $\mathbf{Q} - \mathbf{P}$.
- (ii) Draw $(-1)\mathbf{P}$ and then draw $\mathbf{Q} + (-1)\mathbf{P}$.

Problem 6. Let $\mathbf{P} = (a_1, a_2, a_3)$ and $\mathbf{Q} = (b_1, b_2, b_3)$.

- (i) Find the coordinates $\mathbf{Q} - \mathbf{P} = (\quad , \quad , \quad)$ of the vector you would need to add to $\mathbf{P}$ to get $\mathbf{Q}$.
- (iv) Show how to compute the coordinates $\mathbf{Q} + (-1)\mathbf{P} = (\quad , \quad , \quad)$.

Problem 7. Let $\mathbf{P}$ and $\mathbf{Q}$ be any two vectors. Find a description for the line $\overline{\mathbf{PQ}}$ through $\mathbf{P}$ and $\mathbf{Q}$ by identifying the vector pointing from $\mathbf{P}$ to $\mathbf{Q}$ and adding scalar multiples of it to a single point on it.

One immediate application of these ideas arises when we solve a system of linear equations. For example, suppose we want a good description of the set of all solutions x , y and z to the system of simultaneous linear equations

$$\begin{array}{rclcl} 3x & - & 2y & - & z & = & 0 & [1] \\ 2x & + & y & + & z & = & 10 & [2] \\ x & + & 4y & + & 3z & = & 20 & [3] \end{array}$$

If we think of a solution to this system as a point (x, y, z) , then we are asking for a geometric description of the set $\mathbb{S}$ of all of its solutions.

The important observation to make is that we can transform a system of linear equations into a new system that has exactly the same solutions by changing just one of the equations in one of two ways:

- (i) add a multiple of another equation to it or
- (ii) multiply it by a non-zero number.

By a sequence of these kind of transformations, we can transform the system $\{[1], [2], [3]\}$ into an equivalent system $\{[10], [11]\}$ whose solutions are immediately apparent. Here is a list of the steps we perform.

[4] is obtained by adding -3 times [3] to [1];

[5] is obtained by adding -2 times [3] to [2];

[6] is obtained by multiplying [4] by $-\frac{1}{2}$;

[7] is obtained by adding [6] to [5];

[8] is obtained by multiplying [6] by $\frac{1}{7}$;

[9] is obtained by adding -4 times [8] to [3].

We can eliminate [7] since it is satisfied by every point (x, y, z) and therefore contributes nothing to the solution. We obtain [10] and [11] by solving [8] and [9] for y and x .

$$0x - 14y - 10z = -60 \quad [4]$$

$$0x - 7y - 5z = -30 \quad [5]$$

$$x + 4y + 3z = 20 \quad [3]$$

$$0x - 7y + 5z = 30 \quad [6]$$

$$0x - 0y - 0z = 0 \quad [7]$$

$$x + 4y + 3z = 20 \quad [3]$$

$$0x - y + \frac{5}{7}z = \frac{30}{7} \quad [8]$$

$$x + 4y + 3z = 20 \quad [3]$$

$$0x + y + \frac{5}{7}z = \frac{30}{7} \quad [8]$$

$$x + 0y + \frac{1}{7}z = \frac{20}{7} \quad [9]$$

$$y = \frac{30-5z}{7} \quad [10]$$

$$x = \frac{20-z}{7} \quad [11]$$

The solutions to the system $\{[10], [11]\}$ are obtained by choosing just any value for z and taking $x = \frac{20-z}{7}$ and $y = \frac{30-5z}{7}$. If we let $\mathbf{P} := (\frac{20}{7}, \frac{30}{7}, 0)$ and $\mathbf{Q} := (\frac{-1}{7}, \frac{-5}{7}, 1)$, then the solution set $\mathbb{S}$ is given by

$$\begin{aligned} \mathbb{S} &= \{(\frac{20-z}{7}, \frac{30-5z}{7}, z) \mid z \in \mathbb{R}\} \\ &= \{(\frac{20}{7}, \frac{30}{7}, 0) + z(\frac{-1}{7}, \frac{-5}{7}, 1) \mid z \in \mathbb{R}\} = \{\mathbf{P} + z\mathbf{Q} \mid z \in \mathbb{R}\}. \end{aligned}$$

From Problem 7 we see that $\mathbb{S}$ is the line through $\mathbf{P} = (\frac{20}{7}, \frac{30}{7}, 0)$ in the direction of $\mathbf{Q} = (\frac{-1}{7}, \frac{-5}{7}, 1)$.

Solve each of the following systems in the same way, and give a geometric description of the solution set. Be sure to substitute your solutions into the original equations to see if they are correct.

Problem 8. $4x - y + 3z = 5$ and $3x - y + 2z = 7$.

Problem 9. $7x - 11y - 2z = 3$ and $8x - 2y + 3z = 1$.

Problem 10. $3x - 5y + z = 2$ and $-6x + 10y - 2z = 3$.

We say that a vector $\mathbf{P}$ is a **linear combination** of vectors $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ if there are numbers $t_1, t_2, \dots, t_n$ such that

$$\mathbf{P} = t_1\mathbf{P}_1 + t_2\mathbf{P}_2 + \cdots + t_n\mathbf{P}_n.$$

The **span** of the set $\mathbb{B} := \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n\}$, denoted by $\text{Span}\mathbb{B}$, consists of all vectors that are a linear combination of $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$. In symbols,

$$\text{Span}\mathbb{B} := \{t_1\mathbf{P}_1 + t_2\mathbf{P}_2 + \cdots + t_n\mathbf{P}_n \mid t_1, t_2, \dots, t_n \in \mathbb{R}\}.$$

We denote the set of all vectors as $\mathbb{R}^3$, called Euclidean 3-Space. If we think of a set $\mathbb{B}$ of vectors in $\mathbb{R}^3$ as a set of points in space, then $\text{Span}\mathbb{B}$ is a new set of points forming a larger subset of $\mathbb{R}^3$.

Problem 11. For each of the following sets $\mathbb{B}$ of vectors, give a geometric description of $\text{Span}\mathbb{B}$.

1. $\mathbb{B} = \{(0, 1, 0)\}$
2. $\mathbb{B} = \{(5, -2, 17)\}$
3. $\mathbb{B} = \{(0, 0, 0)\}$
4. $\mathbb{B} = \{(1, 0, 0), (0, 0, 1)\}$
5. $\mathbb{B} = \{-6, -3, 9), (4, 2, -6)\}$
6. $\mathbb{B} = \{(4, -3, 7), (1, 5, 3)\}$
7. $\mathbb{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$
8. $\mathbb{B} = \{(1, 0, 0), (3, 0, 3), (0, 0, 1)\}$
9. $\mathbb{B} = \{(-1, 2, 3), (0, 0, 0), (2, -4, -6)\}$

A set $\mathbb{B}$ of vectors is said to **span** $\mathbb{R}^3$ if $\text{Span}\mathbb{B} = \mathbb{R}^3$; that is, if every vector is a linear combination of the vectors in $\mathbb{B}$. A central problem for us will be that of finding out when a set $\mathbb{B}$ spans $\mathbb{R}^3$.

For each of the following four problems, decide whether or not $\{\mathbf{P}, \mathbf{Q}, \mathbf{R}\}$ spans $\mathbb{R}^3$. To show that it does, you will need to show how an arbitrary point $\mathbf{A} := (a, b, c)$ can be

expressed as a linear combination of $\{\mathbf{P}, \mathbf{Q}, \mathbf{R}\}$, that is, *for every choice of $\mathbf{A}$ there exist x , y and z such that*

$$\mathbf{A} = x\mathbf{P} + y\mathbf{Q} + z\mathbf{R}.$$

To show that it does not, you need to exhibit a particular vector $\mathbf{A}$ that is not in its span, that is, *for some choice of $\mathbf{A}$ there do not exist x , y and z such that*

$$\mathbf{A} = x\mathbf{P} + y\mathbf{Q} + z\mathbf{R}.$$

Problem 12. Let $\mathbf{P} = (1, 0, 0)$, $\mathbf{Q} = (0, 5, 0)$, $\mathbf{R} = (0, 0, -2)$.

Problem 13. Let $\mathbf{P} = (1, 0, 1)$, $\mathbf{Q} = (0, 1, 0)$, $\mathbf{R} = (1, 1, 1)$.

Problem 14. Let $\mathbf{P} = (2, 0, 1)$, $\mathbf{Q} = (1, -1, 1)$, $\mathbf{R} = (0, 3, 2)$.

Problem 15. Let $\mathbf{P} = (1, -1, 2)$, $\mathbf{Q} = (3, 1, 5)$, $\mathbf{R} = (3, 5, 4)$.

Problem 16. Of the previous four problems, look at the ones which do span $\mathbb{R}^3$. For each of these, is there more than one choice of x , y and z or is the choice unique?

A set $\mathbb{B}$ of vectors is called a **basis** for $\mathbb{R}^3$ if every vector can be written in one and only one way as a linear combination of vectors in $\mathbb{B}$.

Problem 17. Let $\mathbf{I} := (1, 0, 0)$, $\mathbf{J} := (0, 1, 0)$, $\mathbf{K} := (0, 0, 1)$. Show that $\mathbb{B} := \{\mathbf{I}, \mathbf{J}, \mathbf{K}\}$ is a basis for $\mathbb{R}^3$.

The basis $\{\mathbf{I}, \mathbf{J}, \mathbf{K}\}$ is called the **standard basis** for $\mathbb{R}^3$.

Given three points $\mathbf{P} = (a_1, a_2, a_3)$, $\mathbf{Q} = (b_1, b_2, b_3)$ and $\mathbf{R} = (c_1, c_2, c_3)$, is there some simple way to know whether or not they span $\mathbb{R}^3$? It turns out that there is, and that it can be established using the ideas we have developed here. We will not prove this fact now, but will simply quote the result.

We define the **determinant** of $\mathbf{P}, \mathbf{Q}$ and $\mathbf{R}$ to be the number

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1$$

$$\begin{array}{ccccccc} \swarrow & \swarrow & \swarrow & \searrow & \searrow & \searrow & \\ - & - & - & + & + & + & \end{array}$$

Theorem Vectors $\mathbf{P}, \mathbf{Q}$ and $\mathbf{R}$ span $\mathbb{R}^3$ if and only if their determinant is not zero.

Problem 18. Test this theorem by looking back at Problems 12, 13, 14, 15 and 17, and computing the determinant of $\mathbf{P}, \mathbf{Q}$ and $\mathbf{R}$ in each case. Show that the determinant is zero for the ones that do not span $\mathbb{R}^3$ and that the determinant is not zero for the ones that do span $\mathbb{R}^3$.

Chapter 2

Linear Spaces

A **linear space** is a set $\mathbb{L}$ of objects called **points** such that, for all $\mathbf{P}$ and $\mathbf{Q}$ in $\mathbb{L}$ and every real number a , there are

- a unique point $\mathbf{P} + \mathbf{Q}$ in $\mathbb{L}$ called the **sum** of $\mathbf{P}$ and $\mathbf{Q}$ and
- a unique point $a\mathbf{P}$ in $\mathbb{L}$ called the **scalar product** of a and $\mathbf{P}$

such that the following axioms hold.

[Addition]

- 1.) $\mathbf{P} + \mathbf{Q} = \mathbf{Q} + \mathbf{P}$ for all points $\mathbf{P}$ and $\mathbf{Q}$,
- 2.) $\mathbf{P} + (\mathbf{Q} + \mathbf{R}) = (\mathbf{P} + \mathbf{Q}) + \mathbf{R}$ for all points $\mathbf{P}, \mathbf{Q}$ and $\mathbf{R}$, and
- 3.) there is a point $\mathbf{0}$ such that $\mathbf{P} + \mathbf{0} = \mathbf{P}$ for every point $\mathbf{P}$.

[Scalar Product] For all points $\mathbf{P}$ and $\mathbf{Q}$ and all real numbers a and b ,

- 4.) $a(\mathbf{P} + \mathbf{Q}) = a\mathbf{P} + a\mathbf{Q}$,
- 5.) $(a + b)\mathbf{P} = a\mathbf{P} + b\mathbf{P}$,
- 6.) $a(b\mathbf{P}) = (ab)\mathbf{P}$ and
- 7.) $a\mathbf{P} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{P} = \mathbf{0}$.

Theorem 19. For every point $\mathbf{Q}$ we have that $\mathbf{Q} + (-1)\mathbf{Q} = \mathbf{0}$.

[Hint: To prove this theorem, apply Axiom 7 taking $a := 1$ and $\mathbf{P} := \mathbf{Q} + (-1)\mathbf{Q}$. Be sure to quote each axiom that you use.]

The point $(-1)\mathbf{Q}$ is usually denoted by $-\mathbf{Q}$ and is called the **additive inverse** of $\mathbf{Q}$. Thus $\mathbf{Q} + (-\mathbf{Q}) = (-\mathbf{Q}) + \mathbf{Q} = \mathbf{0}$ for every point $\mathbf{Q}$.

Theorem 20. For every pair of points $\mathbf{P}$ and $\mathbf{Q}$, there is a unique point $\mathbf{X}$ such that $\mathbf{Q} + \mathbf{X} = \mathbf{P}$; namely, $\mathbf{X} = \mathbf{P} + (-1)\mathbf{Q}$.

We define $\mathbf{P} - \mathbf{Q}$ to be the point $\mathbf{P} + (-\mathbf{Q})$ and call it the **difference** between $\mathbf{P}$ and $\mathbf{Q}$. Thus $\mathbf{Q} + (\mathbf{P} - \mathbf{Q}) = \mathbf{P}$.

Theorem 21. For every pair of points $\mathbf{P}$ and $\mathbf{Q}$ and all real numbers a , we have $a(\mathbf{P} - \mathbf{Q}) = a\mathbf{P} - a\mathbf{Q}$.

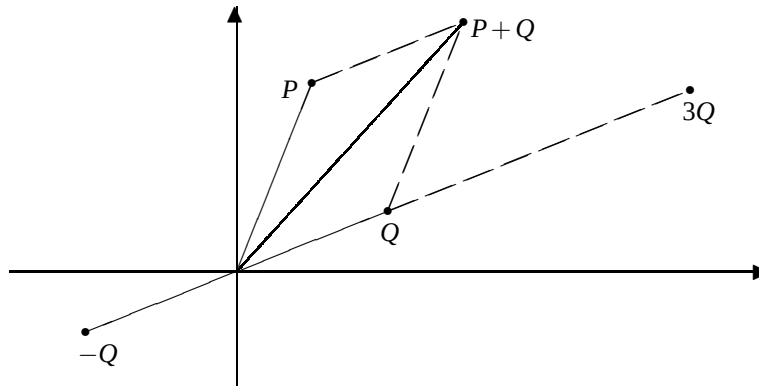
Theorem 22. For every point $\mathbf{P}$ we have $1\mathbf{P} = \mathbf{P}$.

To define a particular linear space, we must specify what its points are, what the sum of two points is and what the scalar product of a number and a point is. Check that each of the following examples is a linear space by verifying Axioms 1 to 7.

1. The space $\mathbb{R}^2$ consists of all ordered pairs of real numbers. If $\mathbf{P} = (a, b)$ and $\mathbf{Q} = (c, d)$, then $\mathbf{P} + \mathbf{Q}$ and $r\mathbf{P}$ are defined as

$$\begin{aligned}(a, b) + (c, d) &:= (a + c, b + d); \\ r(a, b) &:= (ra, rb)\end{aligned}$$

where r is a real number.



2. More generally, for each positive integer n , the space $\mathbb{R}^n$ consists of all ordered n -tuples of real numbers. If $\mathbf{P} = (a_1, \dots, a_n)$ and $\mathbf{Q} = (c_1, \dots, c_n)$, then $\mathbf{P} + \mathbf{Q}$ and $r\mathbf{P}$ are defined as

$$\begin{aligned}(a_1, \dots, a_n) + (c_1, \dots, c_n) &:= (a_1 + c_1, \dots, a_n + c_n); \\ r(a_1, \dots, a_n) &:= (ra_1, \dots, ra_n).\end{aligned}$$

where r is a real number.

3. A real valued function $\mathbf{f}$ is **continuous** if, for each a in its domain, $\lim_{x \rightarrow a} \mathbf{f}(x) = \mathbf{f}(a)$. Let $\mathbb{C}[0, 1]$ denote the set of continuous real valued functions whose domain is the closed interval $[0, 1]$. For $\mathbf{f}, \mathbf{g} \in \mathbb{C}[0, 1]$ and a real number r , we define $\mathbf{f} + \mathbf{g}$ and $r\mathbf{f}$ as

$$\begin{aligned}(\mathbf{f} + \mathbf{g})(x) &:= \mathbf{f}(x) + \mathbf{g}(x); \\ (r\mathbf{f})(x) &:= r(\mathbf{f}(x))\end{aligned}$$

for each $x \in [0, 1]$. From calculus, we know that $\mathbf{f} + \mathbf{g}$ and $r\mathbf{f}$ are continuous if $\mathbf{f}$ and $\mathbf{g}$ are continuous.

4. For each positive integer n , let $\mathbb{P}_n[0, 1]$ denote the set of all $\mathbf{f} \in \mathbb{C}[0, 1]$ such that $\mathbf{f}$ is a polynomial function of degree less than n . Equivalently, $\mathbf{f} \in \mathbb{P}_n[0, 1]$ if there are real numbers $a_0, a_1, \dots, a_{n-1}$ such that, for all $x \in [0, 1]$,

$$\mathbf{f}(x) := a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_{n-1}x^{n-1}.$$

Addition and scalar multiplication are both defined as in $\mathbb{C}[0, 1]$.

A subset $\mathbb{M}$ of a linear space $\mathbb{L}$ is called a **subspace** of $\mathbb{L}$ if the following conditions hold.

- (i) $\mathbb{M}$ is **closed under addition**, that is, $\mathbf{P} + \mathbf{Q} \in \mathbb{M}$ whenever $\mathbf{P}$ and $\mathbf{Q}$ are in $\mathbb{M}$.
- (ii) $\mathbb{M}$ is **closed under scalar multiplication**, that is, $a\mathbf{P} \in \mathbb{M}$ whenever $\mathbf{P}$ is in $\mathbb{M}$ and a is a number.
- (iii) $\mathbb{M}$ is itself a linear space, that is, it also satisfies Axioms 1 to 7.

According to this definition, it is necessary to check 9 conditions to verify that $\mathbb{M}$ is a linear space. Using the fact that $\mathbb{M}$ is contained in a set $\mathbb{L}$ that we already know to be a linear space, we find that much less is necessary.

Theorem 23. *A subset $\mathbb{M}$ of a linear space $\mathbb{L}$ is a subspace of $\mathbb{L}$ if and only if $\mathbf{0} \in \mathbb{M}$ and $\mathbb{M}$ is closed under both addition and scalar multiplication.*

Theorem 24. *Let $\mathbb{M}$ be a subset of a linear space $\mathbb{L}$. Then $\mathbb{M}$ is a subspace of $\mathbb{L}$ if and only if $\mathbf{0} \in \mathbb{M}$ and, for every $\mathbf{P}$ and $\mathbf{Q}$ in $\mathbb{M}$ and every pair a and b of numbers, $a\mathbf{P} + b\mathbf{Q}$ is also in $\mathbb{M}$.*

A wealth of new and interesting linear spaces arise as subspaces of familiar linear spaces.

Problem 25. *For each of the following choices of $\mathbb{M}$, either verify that $\mathbb{M}$ is a subspace of the given space or show that it fails to satisfy one of the defining properties of a subspace.*

- (i) $\mathbb{M}$ consists of all $(x, y) \in \mathbb{R}^2$ satisfying $5x - 3y = 0$.
- (ii) $\mathbb{M}$ consists of all $(x, y) \in \mathbb{R}^2$ satisfying $5x - 3y = 4$.
- (iii) $\mathbb{M}$ consists of all $(x, y, z) \in \mathbb{R}^3$ for which $z = 0$.
- (iv) $\mathbb{M}$ consists of all $(x, y, z) \in \mathbb{R}^3$ for which $z \geq 0$.
- (v) $\mathbb{M}$ consists of all differentiable functions in $\mathbb{C}[0, 1]$.
- (vi) $\mathbb{M}$ consists of all functions $\mathbf{f}$ in $\mathbb{C}[0, 1]$ such that $3\mathbf{f}'' + 5\mathbf{f}' - 2\mathbf{f} = \mathbf{0}$.
- (vii) $\mathbb{M}$ consists of all functions $\mathbf{f}$ in $\mathbb{C}[0, 1]$ such that $3\mathbf{f}'' + 5\mathbf{f}' - 2\mathbf{f} = \mathbf{2}$.
- (viii) $\mathbb{M}$ consists of all functions $\mathbf{f}$ in $\mathbb{C}[0, 1]$ such that $\mathbf{f}(\frac{1}{4}) = 0$.
- (ix) $\mathbb{M}$ consists of all functions $\mathbf{f}$ in $\mathbb{C}[0, 1]$ such that $\mathbf{f}(x)$ is rational for all $x \in [0, 1]$.

Viewing a linear space algebraically has one immediate advantage. Some points can be generated as sums of products of other points, thereby making them in a sense redundant in the presence of those other points. To make this idea precise, we say that a point $\mathbf{P}$ is a **linear combination** of a finite set $\mathbb{B} := \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n\}$ of points if there exist numbers $c_1, c_2, \dots, c_n$ such that

$$\mathbf{P} = c_1\mathbf{P}_1 + c_2\mathbf{P}_2 + \dots + c_n\mathbf{P}_n.$$

The set of all points that are linear combinations of $\mathbb{B}$ is denoted by $\text{Span}(\mathbb{B})$ and is called the **span** of $\mathbb{B}$. If $\mathbb{B} = \emptyset$ is empty, we define $\text{Span}(\mathbb{B}) = \{\mathbf{0}\}$. If $\text{Span}(\mathbb{B}) = \mathbb{L}$ we say that $\mathbb{B}$ **spans** $\mathbb{L}$.

Problem 26. Find two points of $\mathbb{R}^2$ that span $\mathbb{R}^2$. Now find a different pair of points of $\mathbb{R}^2$ that also span $\mathbb{R}^2$.

Problem 27. Find a finite set of points of $\mathbb{R}^n$ that span $\mathbb{R}^n$. Now find a different set of points of $\mathbb{R}^n$ that also span $\mathbb{R}^n$.

Problem 28. Find a finite set of points of $\mathbb{P}_n[0, 1]$ that spans $\mathbb{P}_n[0, 1]$.

We can extend the definition of “span” to an infinite subset $\mathbb{B}$ of a linear space $\mathbb{L}$ by defining $\text{Span}(\mathbb{B})$ to be the set of all linear combinations of a finite subsets of $\mathbb{B}$. For example, let $\mathbb{P}[0, 1]$ denote the linear space of all polynomial functions with domain $[0, 1]$, that is, the union of all $\mathbb{P}_n[0, 1]$ where $n > 0$.

Problem 29. Find an infinite set of points of $\mathbb{P}[0, 1]$ that spans $\mathbb{P}[0, 1]$. Does $\mathbb{P}[0, 1]$ have a finite spanning set?

Problem 30. Is there a finite set of points of $\mathbb{C}[0, 1]$ that spans $\mathbb{C}[0, 1]$?

Theorem 31. If $\mathbb{B}$ is a subset of a linear space $\mathbb{L}$, then $\text{Span}(\mathbb{B})$ is a subspace of $\mathbb{L}$. Moreover, $\text{Span}(\mathbb{B})$ is the **smallest** subspace of $\mathbb{L}$ containing $\mathbb{B}$ in the sense that, if $\mathbb{M}$ is any subspace of $\mathbb{L}$ containing $\mathbb{B}$, then $\text{Span}(\mathbb{B})$ is contained in $\mathbb{M}$.

We say that a subset $\mathbb{B}$ of a linear space $\mathbb{L}$ is **linearly independent** if every point $\mathbf{Q}$ in $\text{Span}(\mathbb{B})$ can be written in only one way as a linear combination of $\mathbb{B}$, that is, for $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n \in \mathbb{B}$ and numbers $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ we have that

$$a_1\mathbf{P}_1 + a_2\mathbf{P}_2 + \dots + a_n\mathbf{P}_n = b_1\mathbf{P}_1 + b_2\mathbf{P}_2 + \dots + b_n\mathbf{P}_n$$

implies that $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$. (See Problem 16.) We say that $\mathbb{B}$ is **linearly dependent** if it is not linearly independent, that is, if some point in $\text{Span}(\mathbb{B})$ can be written in two different ways as a linear combination of $\mathbb{B}$.

Theorem 32. If $\mathbb{B}$ is a subset of a linear space $\mathbb{L}$ and $\mathbf{0} \in \mathbb{B}$, then $\mathbb{B}$ is linearly dependent.

Theorem 33. If $\mathbb{B}$ is a subset of a linear space $\mathbb{L}$, then $\mathbb{B}$ is linearly independent if and only if for all $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ in $\mathbb{B}$ and all numbers $a_1, a_2, \dots, a_n$

$$a_1\mathbf{P}_1 + a_2\mathbf{P}_2 + \dots + a_n\mathbf{P}_n = \mathbf{0} \quad \text{implies} \quad a_1 = a_2 = \dots = 0.$$

Theorem 34. Let $\mathbb{B}$ be a subset of a linear space $\mathbb{L}$ containing more than one point. Then $\mathbb{B}$ is linearly dependent if and only if some point of $\mathbb{B}$ is a linear combination of the other points of $\mathbb{B}$.

Problem 35. How many points can be in a linearly independent subset of $\mathbb{R}^2$? How can we identify linearly independent subsets of $\mathbb{R}^2$ geometrically?

Problem 36. Does $\mathbb{C}[0, 1]$ have an infinite linearly independent subset? [Note: You don't need formulas for functions; just graphs will do.]

The ideas of linear independence and spanning sets combine to give us one of the central concepts of linear algebra. A subset $\mathbb{B}$ of a linear space $\mathbb{L}$ is a **basis** for $\mathbb{L}$ if each point of $\mathbb{L}$ can be expressed in one and only one way as a linear combination of points of $\mathbb{B}$. In our terminology, this is the same as saying that $\mathbb{B}$ is a **basis** for $\mathbb{L}$ if it is linearly independent and it spans $\mathbb{L}$.

Problem 37. Give a geometric description of the subsets $\mathbb{B}$ of $\mathbb{R}^2$ and $\mathbb{R}^3$ that form bases. Do all bases for one of these spaces have the same number of points?

Problem 38. Find a basis for each of the following spaces.

$$(i)\mathbb{R}^7 \quad (ii)\mathbb{P}_9[0, 1] \quad (iii)\mathbb{P}[0, 1].$$

Lemma 39. Suppose $\mathbb{B}$ is a linearly independent subset of $\mathbb{L}$ and P is a point of $\mathbb{L}$ not in $\text{Span}(\mathbb{B})$. Then $\mathbb{B} \cup \{P\}$ is also linearly independent.

Theorem 40. $\mathbb{B}$ is a basis for $\mathbb{L}$ if and only if it is a maximal linearly independent subset of $\mathbb{L}$, that is, it is linearly independent but is not a proper subset of any other linearly independent set.

Lemma 41. Suppose $\mathbb{B}$ spans $\mathbb{L}$ and P is a point of $\mathbb{B}$ such that $P \in \text{Span}(\mathbb{B} - \{P\})$. Then $\mathbb{B} - \{P\}$ also spans $\mathbb{L}$.

Theorem 42. $\mathbb{B}$ is a basis for $\mathbb{L}$ if and only if it is a minimal spanning set for $\mathbb{L}$, that is, it spans $\mathbb{L}$ but no proper subset of $\mathbb{B}$ spans $\mathbb{L}$.

We know that $\mathbb{R}^{14}$ has a basis of 14 points. Do you think that there might also be a basis for $\mathbb{R}^{14}$ with 13 points, or maybe 17 points? We will now see that these kind of strange things can never happen.

Replacement Lemma 43. Suppose that $\mathbb{B}$ spans $\mathbb{L}$, that $Q \in \mathbb{L}$ and that P is a point of $\mathbb{B}$ such that, when Q is written as a linear combination of points of $\mathbb{B}$, the coefficient of P is not zero. Let $\mathbb{B}'$ be the set obtained from $\mathbb{B}$ by replacing P with Q . Then $\mathbb{B}'$ also spans $\mathbb{L}$.

Fundamental Lemma 44. Considering only finite subsets of $\mathbb{L}$, no linearly independent set has more points than a spanning set.

Linear Independence Theorem 45. Suppose that $\mathbb{L}$ has a basis with a finite number n of points. Then the following are all true.

- (i) No linearly independent set contains more than n points.
- (ii) Every linearly independent set with n points is a basis.
- (iii) Every linearly independent set is contained in a basis.

Spanning Theorem 46. Suppose that $\mathbb{L}$ has a basis with a finite number n of points. Then the following are all true.

- (i) No spanning set contains fewer than n points.
- (ii) Every spanning set with n points is a basis.
- (iii) Every spanning set contains a basis.

Basis Theorem 47. If $\mathbb{L}$ has a basis with a finite number n of points, then every basis for $\mathbb{L}$ has n points.

Problem 48. Show that no finite set spans $\mathbb{C}[0, 1]$.

If a linear space $\mathbb{L}$ has a finite basis, then the **dimension** of $\mathbb{L}$ is the number n of points in every basis for $\mathbb{L}$. The following theorem can be used to find a basis for a space $\mathbb{L}$ if we are given a finite spanning set for $\mathbb{L}$. Use the Replacement Lemma to prove it.

Theorem 49. Let $\mathbb{B}$ be a spanning set for $\mathbb{L}$ and let $\mathbf{P}$ be in $\mathbb{B}$. Assume that $\mathbf{Q}$ is obtained from $\mathbf{P}$ either

- (i) by adding to $\mathbf{P}$ a linear combination of the remaining elements of $\mathbb{B}$ or
- (ii) by multiplying $\mathbf{P}$ by a nonzero number.

Let $\mathbb{B}'$ be obtained from $\mathbb{B}$ by replacing $\mathbf{P}$ with $\mathbf{Q}$. Then $\mathbb{B}'$ also spans $\mathbb{L}$.

This theorem will give us an algorithm to can find a basis for the subspace of $\mathbf{R}^n$ spanned by any given finite set of points. An m -by- n **matrix** M has m rows, each a point of $\mathbf{R}^n$. The matrix M is said to be **row reduced** if the first nonzero entry in each row is a 1, and the entries above and below that 1 are all 0.

Lemma 50. The nonzero rows of a reduced matrix are linearly independent.

Theorem 51. Let S be a set of m points of $\mathbf{R}^n$ and let $\mathbb{L} = \text{Span}(S)$. Let M be the m -by- n matrix obtained by stacking the points of S in a column and let M' be the reduced matrix obtained from M by repeated applications of Theorem 49 (called **Gaussian elimination**). Then the nonzero rows of M' form a basis for $\mathbb{L}$.

Problem 52. Find a basis for the subspace of $\mathbf{R}^5$ spanned by the points $(0, 5, 1, 2, 3)$, $(1, 0, 1, 6, 4)$, $(0, 0, 1, 2, 3)$ and $(1, 5, 2, 8, 7)$.

Problem 53. Find a basis for the subspace of $\mathbf{R}^3$ spanned by the points $(8, 3, 5)$, $(5, 1, 9)$, $(1, 3, -1)$ and $(2, -1, 5)$.

Problem 54. Find a basis for the subspace of $\mathbf{R}^4$ spanned by the points $(6, -2, 1, -3)$, $(-5, 2, 0, 1)$, $(1, 0, 1, -2)$ and $(3, -2, -2, 3)$.

How can we describe the solution set $\mathbb{S}$ of the single linear equation

$$4x - y + 3z = 5?$$

We can answer this exactly as we did the systems of linear equations in Chapter 1 by noticing that any values of y and z will determine a unique value of x such that (x, y, z) is a solution.

$$\begin{aligned}\mathbb{S} &= \{(x, y, z) \mid x = \frac{5+y-3z}{4}\} \\ &= \{(\frac{5+y-3z}{4}, y, z) \mid y, z \in \mathbb{R}\} \\ &= \{(\frac{5}{4}, 0, 0) + (\frac{y}{4}, y, 0) + (\frac{-3z}{4}, 0, z) \mid y, z \in \mathbb{R}\} \\ &= \{(\frac{5}{4}, 0, 0) + y(\frac{1}{4}, 1, 0) + z(\frac{-3}{4}, 0, 1) \mid y, z \in \mathbb{R}\}.\end{aligned}$$

If we take $\mathbf{Q} := (\frac{5}{4}, 0, 0)$, $\mathbf{P}_1 := (\frac{1}{4}, 1, 0)$ and $\mathbf{P}_2 := (\frac{-3}{4}, 0, 1)$, then this tells us that $\mathbb{S}$ is what we get when we add $\mathbf{Q}$ to each point in the span of $\{\mathbf{P}_1, \mathbf{P}_2\}$. We write this as

$$\mathbb{S} = \mathbf{Q} + \text{Span}\{\mathbf{P}_1, \mathbf{P}_2\}.$$

Applying Theorem 33, we can verify that $\mathbb{B} := \{\mathbf{P}_1, \mathbf{P}_2\}$ is linearly independent and is therefore a basis for the linear subspace $\mathbb{L} := \text{Span}\{\mathbf{P}_1, \mathbf{P}_2\}$ of $\mathbb{R}^3$. Since $\mathbb{L}$ is a 2-dimensional subspace of $\mathbb{R}^3$, it is a plane that looks just like $\mathbb{R}^2$. Then $\mathbb{S}$ is obtained from $\mathbb{L}$ by moving it $\frac{5}{4}$ units in the positive x direction. Thus $\mathbb{S}$ is also a plane (but not a subspace).

As a check (though not a proof) of the correctness of our work, we can test one point $\mathbf{P}$ in $\mathbb{S}$ to see if it is a solution to the original system. We pick any values for y and z ; say $y = 6$ and $z = 1$. This gives us

$$\mathbf{P} = (\frac{5}{4}, 0, 0) + 6(\frac{1}{4}, 1, 0) + 1(\frac{-3}{4}, 0, 1) = (2, 6, 1).$$

Putting $\mathbf{P}$ into the original equation, we get $4x - 6y + 3z = 4 \cdot 2 - 6 \cdot 6 + 3 \cdot 1 = 5$, so it is indeed a solution.

Find a description of the solution set of each system of linear equations below by carrying out the following steps.

- (i) Use Gaussian elimination to find the solution set $\mathbb{S}$ as you did in Chapter 1.
- (ii) Find a point $\mathbf{Q}$ and a set of points $\mathbb{B} := \{\mathbf{P}_1, \mathbf{P}_2, \dots\}$ so that $\mathbb{S} = \mathbf{Q} + \text{Span}\mathbb{B}$.
- (iii) Show that $\mathbb{B}$ is a basis for $\mathbb{L} := \text{Span}\mathbb{B}$. What is the dimension of the space $\mathbb{L}$?
- (iv) Describe $\mathbb{S}$ as looking like either a line ($\mathbb{R}^1$), a plane ($\mathbb{R}^2$), 3-space ($\mathbb{R}^3$), etc.
- (v) Compute one point $\mathbf{A} = (a, b, c, \dots)$ that is in $\mathbb{S}$. Check your work by verifying that $\mathbf{A}$ is a solution to each of the original equations.

Problem 55.

$$3x + y - 7z - 4u = 6$$

Problem 56.

$$\begin{array}{ccccccccc} x & + & 2y & + & z & + & 3u & + & 6v & = & 7 \\ -x & & & & & & -6u & + & 5v & = & -8 \\ 2x & & & & -z & + & 12u & - & 15v & = & 10 \end{array}$$

Problem 57.

$$\begin{array}{rcccccccl} x & + & y & - & z & - & 6u & = & 4 \\ 2x & + & 5y & & & - & 14u & = & 26 \\ 3x & + & 7y & - & z & - & 24u & = & 36 \end{array}$$

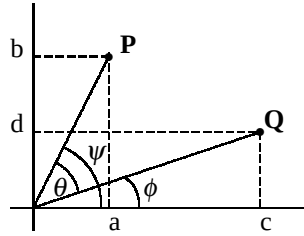
Chapter 3

Inner Product Spaces

The **inner product** (or **dot product**) of two points $\mathbf{P} = (a, b)$ and $\mathbf{Q} = (c, d)$ in $\mathbb{R}^2$ is defined by

$$\mathbf{P} \cdot \mathbf{Q} = (a, b) \cdot (c, d) := ac + bd.$$

To see the significance of this notion, let $\|\mathbf{P}\|$ denote the distance from $\mathbf{P}$ to the origin (or the length of the vector represented by $\mathbf{P}$). From trigonometry, we have



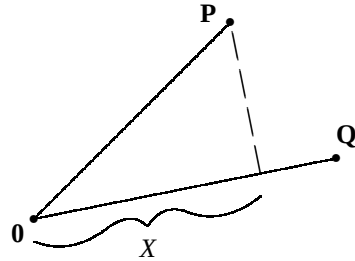
$$\begin{aligned} \cos(\theta) &= \cos(\psi - \phi) = \cos(\psi) \cos(\phi) + \sin(\psi) \sin(\phi) \\ &= \frac{a}{\|\mathbf{P}\|} \frac{c}{\|\mathbf{Q}\|} + \frac{b}{\|\mathbf{P}\|} \frac{d}{\|\mathbf{Q}\|} = \frac{\mathbf{P} \cdot \mathbf{Q}}{\|\mathbf{P}\| \|\mathbf{Q}\|} \end{aligned}$$

Thus

$$\mathbf{P} \cdot \mathbf{Q} = \|\mathbf{P}\| \|\mathbf{Q}\| \cos(\theta).$$

In the diagram on the right, X is called the **component of $\mathbf{P}$ in the direction of $\mathbf{Q}$** . Thus we have

$$X = \|\mathbf{P}\| \cos(\theta) = \frac{\mathbf{P} \cdot \mathbf{Q}}{\|\mathbf{Q}\|}.$$



The space $\mathbb{R}^2$ not only has a structure as a linear space; it also has a geometric structure. The geometry of $\mathbb{R}^2$ comes from the fact that we can talk about the distance between points in $\mathbb{R}^2$. The distance from $\mathbf{P}$ to $\mathbf{Q}$ can be defined as the length $\|\mathbf{P} - \mathbf{Q}\|$, and length can be defined in terms of inner product as $\|\mathbf{P}\| = \sqrt{\mathbf{P} \cdot \mathbf{P}}$. Thus the inner product gives rise to the geometric structure of $\mathbb{R}^2$.

This observation suggests that we look at other linear spaces which have a similar inner product. To do so, we start by stating the essential properties of the inner product. By an **inner product space** we mean a linear space $\mathbb{L}$ in which there is a way to multiply any point $\mathbf{P}$ and any point $\mathbf{Q}$ to obtain a number $\mathbf{P} \cdot \mathbf{Q}$ (called the **inner product** of $\mathbf{P}$ and $\mathbf{Q}$) so that, for all $\mathbf{P}, \mathbf{Q}, \mathbf{R} \in \mathbb{L}$ and all $a \in \mathbb{R}$, the following additional axioms hold.

$$8.) \mathbf{P} \cdot \mathbf{P} \geq 0, \text{ and } \mathbf{P} \cdot \mathbf{P} = 0 \text{ if and only if } \mathbf{P} = \mathbf{0}.$$

$$9.) \mathbf{P} \cdot \mathbf{Q} = \mathbf{Q} \cdot \mathbf{P}$$

$$10.) a(\mathbf{P} \cdot \mathbf{Q}) = (a\mathbf{P}) \cdot \mathbf{Q}$$

$$11.) \mathbf{P} \cdot (\mathbf{Q} + \mathbf{R}) = \mathbf{P} \cdot \mathbf{Q} + \mathbf{P} \cdot \mathbf{R}$$

The **norm** of a point $\mathbf{P}$ is defined to be

$$\|\mathbf{P}\| := \sqrt{\mathbf{P} \cdot \mathbf{P}},$$

and the **distance** from $\mathbf{P}$ to $\mathbf{Q}$ is defined to be

$$\text{distance}(\mathbf{P}, \mathbf{Q}) := \|\mathbf{P} - \mathbf{Q}\|,$$

the norm of the difference. Check that each of the following examples is an inner product space by verifying Axioms 8 to 11.

$$1. \text{ The space } \mathbb{R}^2 \text{ where } (a, b) \cdot (c, d) := ac + bd.$$

$$2. \text{ The space } \mathbb{R}^n \text{ where } (a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_n) := a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

$$3. \text{ The space } \mathbb{C}[0, 1] \text{ where } f \cdot g := \int_0^1 f(x)g(x)dx.$$

$$\textbf{Lemma 58. } (a\mathbf{P} + b\mathbf{Q}) \cdot (c\mathbf{U} + d\mathbf{V}) = (ac)\mathbf{P} \cdot \mathbf{U} + (ad)\mathbf{P} \cdot \mathbf{V} + (bc)\mathbf{Q} \cdot \mathbf{U} + (bd)\mathbf{Q} \cdot \mathbf{V}$$

$$\textbf{Lemma 59. } \|a\mathbf{P}\| = |a|\|\mathbf{P}\| \text{ and } \mathbf{0} \cdot \mathbf{P} = 0.$$

Much of what we will do in general inner product spaces will be motivated by what we see in $\mathbb{R}^2$ and $\mathbb{R}^3$. For example, we found in $\mathbb{R}^2$ that $\mathbf{P} \cdot \mathbf{Q} = \|\mathbf{P}\|\|\mathbf{Q}\|\cos(\theta)$ for every pair of points $\mathbf{P}$ and $\mathbf{Q}$ separated by an angle θ . While we cannot talk about angles between vectors in an arbitrary space inner product space $\mathbb{L}$, one consequence of this fact that is meaningful in $\mathbb{L}$ is also true in $\mathbb{L}$.

Theorem 60. (Cauchy-Schwartz Inequality)

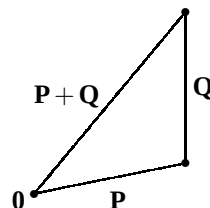
$$|\mathbf{P} \cdot \mathbf{Q}| \leq \|\mathbf{P}\|\|\mathbf{Q}\|$$

[Hint: What does the sign of $f(x) = \|\mathbf{xP} + \mathbf{Q}\|^2$ tell us about $\mathbf{P}$ and $\mathbf{Q}$?

In $\mathbb{R}^2$ we know that the length of one side of a triangle is less than or equal to the sum of the lengths of the other two sides. This fact has an interpretation in an arbitrary inner product space, and we can prove that it is true in every inner product space.

Theorem 61. (Triangle Inequality)

$$\|\mathbf{P} + \mathbf{Q}\| \leq \|\mathbf{P}\| + \|\mathbf{Q}\|$$



We are accustomed to using the standard bases

$$\{\mathbf{i}, \mathbf{j}\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad \{\mathbf{i}, \mathbf{j}, \mathbf{k}\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

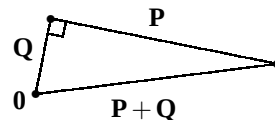
for $\mathbb{B}^2$ and $\mathbb{B}^3$. We will now see how similar standard bases can be constructed for an arbitrary inner product space $\mathbb{L}$, and in the process of doing so see what it is that is special about these bases.

In $\mathbb{B}^2$ the equation $\mathbf{P} \cdot \mathbf{Q} = \|\mathbf{P}\| \|\mathbf{Q}\| \cos(\theta)$ tells us that $\mathbf{P}$ and $\mathbf{Q}$ are orthogonal (perpendicular) if and only if $\mathbf{P} \cdot \mathbf{Q} = 0$. Accordingly, for $\mathbf{P}, \mathbf{Q} \in \mathbb{L}$, we define $\mathbf{P}$ and $\mathbf{Q}$ to be **orthogonal** if $\mathbf{P} \cdot \mathbf{Q} = 0$. A subset $\mathbb{B}$ of $\mathbb{L}$ is **orthogonal** if every pair of distinct points in $\mathbb{B}$ are orthogonal.

Lemma 62. Every orthogonal set $\mathbb{B}$ not containing $\mathbf{0}$ is linearly independent.

Pythagorean Lemma 63. If $\mathbf{P}$ is orthogonal to $\mathbf{Q}$, then

$$\|\mathbf{P}\|^2 + \|\mathbf{Q}\|^2 = \|\mathbf{P} + \mathbf{Q}\|^2.$$



A **unit vector** is a point $\mathbf{U}$ which has norm 1. Points $\mathbf{P}$ and $\mathbf{Q}$ are said to be **in the same direction** if, for some $a > 0$, we have $\mathbf{P} = a\mathbf{Q}$.

Lemma 64. For every point $\mathbf{P}$ other than $\mathbf{0}$, the point $\mathbf{U} := \mathbf{P} / \|\mathbf{P}\|$ is a unit vector in the direction of $\mathbf{P}$.

An **orthonormal set** is a set of unit vectors in $\mathbb{L}$ that forms an orthogonal set. An **orthonormal basis** is an orthonormal set which is also a basis for $\mathbb{L}$. If $\mathbb{B}$ is a basis for $\mathbb{L}$, then every point $\mathbf{P}$ of $\mathbb{L}$ can be expressed uniquely as

$$\mathbf{P} = a_1 \mathbf{P}_1 + a_2 \mathbf{P}_2 + \cdots + a_n \mathbf{P}_n$$

where each $\mathbf{P}_i \in \mathbb{B}$. Given the basis $\mathbb{B}$ and the point $\mathbf{P}$, how can we find the coordinates $a_1, a_2, \dots$ for $\mathbf{P}$ with respect to $\mathbb{B}$? In general this will lead us to solving a large system of simultaneous linear equations. It turns out, in an inner product space, that there is a much more efficient way to find coordinates.

Given a basis $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n\}$ for an inner product space $\mathbb{L}$ and a point $\mathbf{P} \in \mathbb{L}$, we define the numbers $a_{1P}, a_{2P}, \dots, a_{nP}$ as the inner products

$$a_{1P} := \mathbf{P} \cdot \mathbf{P}_1, \quad a_{2P} := \mathbf{P} \cdot \mathbf{P}_2, \quad \dots, \quad a_{nP} := \mathbf{P} \cdot \mathbf{P}_n.$$

For example, consider the standard basis $\mathbb{B} = \{\mathbf{i} = (1, 0), \mathbf{j} = (0, 1)\}$ for $\mathbb{R}^2$. For points $\mathbf{A} = (u, v)$ and $\mathbf{B} = (x, y)$ of $\mathbb{R}^2$ we obtain the inner products

$$a_{1A} := \mathbf{A} \cdot \mathbf{i} = u, \quad a_{2A} := \mathbf{A} \cdot \mathbf{j} = v, \quad a_{1B} := \mathbf{B} \cdot \mathbf{i} = x, \quad a_{2B} := \mathbf{B} \cdot \mathbf{j} = y.$$

Notice that these are exactly the unique coefficients needed to express $\mathbf{A}$ and $\mathbf{B}$ as linear combinations of this basis:

$$\mathbf{A} = a_{1A}\mathbf{i} + a_{2A}\mathbf{j} \quad \text{and} \quad \mathbf{B} = a_{1B}\mathbf{i} + a_{2B}\mathbf{j}.$$

We can also use them to find inner products and norms:

$$\mathbf{A} \cdot \mathbf{B} = a_{1A}a_{1B} + a_{2A}a_{2B} \quad \text{and} \quad \|\mathbf{A}\| = \sqrt{a_{1A}^2 + a_{2A}^2}.$$

Can we do this with other bases as well?

Problem 65. Let $\mathbf{A} = (12, 5)$ and let $\mathbf{B} = (1, 15)$, and consider two bases for $\mathbf{R}^2$:

- (i) $\mathbb{B}_1 = \{\mathbf{P}_1 = (5, 9), \mathbf{P}_2 = (-2, 3)\}$ (an arbitrary basis) and
- (ii) $\mathbb{B}_2 = \{\mathbf{P}_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2}), \mathbf{P}_2 = (\frac{\sqrt{3}}{2}, -\frac{1}{2})\}$ (an orthonormal basis).

For each of these two bases,

- (a) compute the inner products $a_{1A}, a_{2A}, a_{1B}, a_{2B}$ of $\mathbf{A}$ and $\mathbf{B}$ with respect to this basis.
- (b) Is $\mathbf{A} = a_{1A}\mathbf{P}_1 + a_{2A}\mathbf{P}_2$? Is $\mathbf{B} = a_{1B}\mathbf{P}_1 + a_{2B}\mathbf{P}_2$?
- (c) Is $\mathbf{A} \cdot \mathbf{B} = a_{1A}a_{1B} + a_{2A}a_{2B}$?
- (d) Is $\|\mathbf{A}\| = \sqrt{a_{1A}^2 + a_{2A}^2}$?

These experiments lead us to the following theorems.

Theorem 66. Let $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n\}$ be an orthonormal basis for $\mathbb{L}$ and let $\mathbf{P} \in \mathbb{L}$ be a point of $\mathbb{L}$. For $i = 1, 2, \dots, n$, let $a_i = \mathbf{P} \cdot \mathbf{P}_i$. Then $a_1, a_2, \dots$ are the coefficients for $\mathbf{P}$ with respect to $\mathbb{B}$, that is,

$$\mathbf{P} = a_1\mathbf{P}_1 + a_2\mathbf{P}_2 + \dots + a_n\mathbf{P}_n.$$

Theorem 67. Let $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n\}$ be an orthonormal basis for $\mathbb{L}$, let $\mathbf{P} = a_1\mathbf{P}_1 + a_2\mathbf{P}_2 + \dots + a_n\mathbf{P}_n$ and $\mathbf{Q} = b_1\mathbf{P}_1 + b_2\mathbf{P}_2 + \dots + b_n\mathbf{P}_n$. Then

- (i) $\mathbf{P} \cdot \mathbf{Q} = a_1b_1 + a_2b_2 + \dots + a_nb_n$,
- (ii) $\|\mathbf{P}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$.

These nice properties of orthonormal bases lead us to ask two questions about inner product spaces.

- Does every finite-dimensional inner product space have an orthonormal basis?
- If so, is there a practical method to find one?

To answer these questions, let $\mathbb{M}$ be a subspace of an inner product space $\mathbb{L}$ and let $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k\}$ be an orthonormal basis for $\mathbb{M}$. For each point $\mathbf{P} \in \mathbb{L}$, we define the **orthogonal projection of $\mathbf{P}$ in $\mathbb{M}$** to be the point

$$\mathbf{P}_{\mathbb{M}} := a_1\mathbf{P}_1 + a_2\mathbf{P}_2 + \dots + a_k\mathbf{P}_k \quad \text{where} \quad a_i = \mathbf{P} \cdot \mathbf{P}_i \text{ for } i = 1, 2, \dots, k.$$

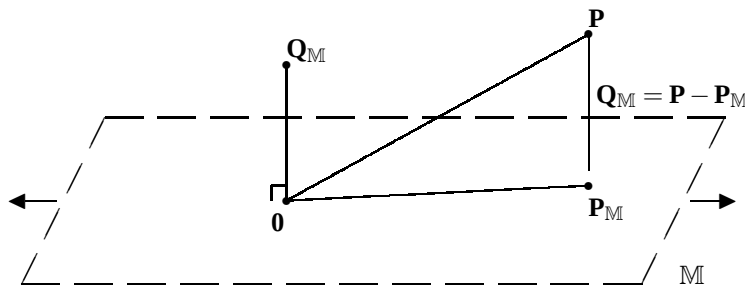
If $\mathbf{P}$ is a point in $\mathbb{M}$, then Theorem 66 tells us that $\mathbf{P}_{\mathbb{M}}$ is $\mathbf{P}$ itself. As another example, consider the orthonormal basis $\mathbb{B} = \{\mathbf{i} = (1, 0, 0), \mathbf{j} = (0, 1, 0)\}$ for a subspace $\mathbb{M}$ of $\mathbb{R}^3$. If $\mathbf{P} = (a, b, c) \in \mathbb{R}^3$, then

$$\mathbf{P}_{\mathbb{M}} = (\mathbf{P} \cdot \mathbf{i}, \mathbf{P} \cdot \mathbf{j}, \mathbf{P} \cdot \mathbf{k}) = (a, b, 0)$$

is the closest point of $\mathbb{M}$ to $\mathbf{P}$.

Lemma 68. Assume that $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k\}$ is an orthonormal basis for a subspace $\mathbb{M}$ of $\mathbb{L}$. Let $\mathbf{P}$ be any point of $\mathbb{L}$. Then

- (i) $\mathbf{Q}_{\mathbb{M}} := \mathbf{P} - \mathbf{P}_{\mathbb{M}}$ is orthogonal to every point of $\mathbb{M}$.
- (ii) $\mathbf{P}_{\mathbb{M}}$ is the closest point of $\mathbb{M}$ to $\mathbf{P}$.



We can now prove an appropriate refinement of Lemma 39 for orthonormal bases.

Lemma 69. Assume that $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k\}$ is an orthonormal set that does not span $\mathbb{L}$. Let $\mathbb{M} := \text{Span}(\mathbb{B})$ and choose any point $\mathbf{P} \notin \mathbb{M}$. Then $\{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k, \mathbf{P}_{k+1}\}$ is also an orthonormal set where $\mathbf{P}_{k+1} := \mathbf{Q}_{\mathbb{M}} / \|\mathbf{Q}_{\mathbb{M}}\|$.

Gram-Schmidt Theorem 70. Let $\mathbb{L}$ be a finite dimensional inner product space. Then every orthonormal subset of $\mathbb{L}$ is contained in an orthonormal basis for $\mathbb{L}$. In particular, $\mathbb{L}$ has an orthonormal basis.

The proof of the Gram-Schmidt Theorem gives us an algorithm to find an orthonormal basis for a finite-dimensional inner product space. This algorithm is called the **Gram-Schmidt Process**.

Problem 71. Illustrate the steps of the Gram-Schmidt Process by finding an orthonormal basis for $\mathbb{R}^3$ which contains a unit vector in the direction of $\mathbf{P} = (1, 2, 2)$.

Using the right orthonormal basis can simplify many tasks. To illustrate this phenomenon we revisit the Cauchy-Schwartz and Triangle Inequalities. Our previous rabbit-out-of-the-hat proof of Cauchy-Schwartz Inequality was correct but rather mysterious. With the right orthonormal basis, we can simply compute both sides of these inequalities and compare the results.

Theorem 72. (Cauchy-Schwartz Revisited)

$$|\mathbf{P} \cdot \mathbf{Q}| \leq \|\mathbf{P}\| \|\mathbf{Q}\|$$

[Hint: Find an orthonormal basis for the subspace $\mathbb{M} := \text{Span}\{\mathbf{P}, \mathbf{Q}\}$ of $\mathbb{L}$ which contains a vector in the direction of $\mathbf{P}$. Then use this basis to compute $|\mathbf{P} \cdot \mathbf{Q}|$ and $\|\mathbf{P}\|\|\mathbf{Q}\|$.]

Theorem 73. (Triangle Inequality Revisited)

$$\|\mathbf{P} + \mathbf{Q}\| \leq \|\mathbf{P}\| + \|\mathbf{Q}\|$$

[Hint: Do as above, but compare $\|\mathbf{P} + \mathbf{Q}\|^2$ with $(\|\mathbf{P}\| + \|\mathbf{Q}\|)^2$.]

Chapter 4

Linear Transformations

Suppose that you go to buy a new car, and that you find two cars for sale of the same make, model, color and price with the same options. They are not literally the same car, but as far as you – the consumer – are concerned, they are indistinguishable. The company that you are buying from may produce thousands of individual cars every year, but only a handful of essentially different kinds. Your task is to choose among the handful of different kinds, not among the thousands of different individual cars.

The same happens with linear spaces and inner product spaces. In order to understand these structures, our task is to recognize the essentially different kinds.

For example, the inner product space $\mathbb{R}^2$ has the standard orthonormal basis $\{\mathbf{i} = (1, 0), \mathbf{j} = (0, 1)\}$. For arbitrary points

$$\mathbf{P} = a\mathbf{i} + b\mathbf{j} \quad \text{and} \quad \mathbf{Q} = c\mathbf{i} + d\mathbf{j}$$

in $\mathbb{R}^2$, addition, scalar multiplication and inner product are given by

$$\mathbf{P} + \mathbf{Q} = (a + c)\mathbf{i} + (b + d)\mathbf{j}, \quad r\mathbf{P} = r a\mathbf{i} + r b\mathbf{j}, \quad \mathbf{P} \cdot \mathbf{Q} = ac + bd.$$

Now consider any other two dimensional inner product space $\mathbb{L}'$. By the Gram-Schmidt Theorem, $\mathbb{L}'$ has an orthonormal basis $\{\mathbf{i}', \mathbf{j}'\}$. For arbitrary points

$$\mathbf{P}' = a\mathbf{i}' + b\mathbf{j}' \quad \text{and} \quad \mathbf{Q}' = c\mathbf{i}' + d\mathbf{j}'$$

in $\mathbb{L}'$, addition, scalar multiplication and inner product are given by

$$\mathbf{P}' + \mathbf{Q}' = (a + c)\mathbf{i}' + (b + d)\mathbf{j}', \quad r\mathbf{P}' = r a\mathbf{i}' + r b\mathbf{j}', \quad \mathbf{P}' \cdot \mathbf{Q}' = ac + bd.$$

What we see is that, although the points of $\mathbb{R}^2$ and $\mathbb{L}'$ may be very different sets of objects, as inner product spaces $\mathbb{R}^2$ and $\mathbb{L}'$ are absolutely indistinguishable. They are like two new red Toyota Corollas with standard transmission, airbags, no AC, and a radio and tape deck that are being sold at the same price. In mathematical terminology which we will define below, we say that $\mathbb{R}^2$ and $\mathbb{L}'$ are **isomorphic** as inner product spaces.

In this sense we think of $\mathbb{R}^2$ as being the *only* two dimensional inner product space. A typical application of this concept was given at the end of the last chapter. Assume we want to prove some theorems about two (linearly independent) points $\mathbf{P}$ and $\mathbf{Q}$ of an arbitrary

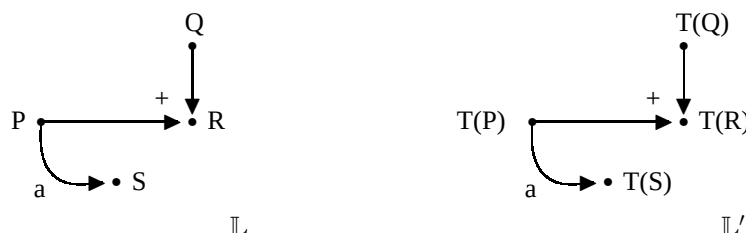
inner product space. By using the fact that the subspace they span is isomorphic to $\mathbb{R}^2$, we could simply paraphrase proofs that the theorems were true in $\mathbb{R}^2$.

The important tool for describing the indistinguishability of $\mathbb{R}^2$ and $\mathbb{L}'$ above is the correspondence

$$a\mathbf{i} + b\mathbf{j} \leftrightarrow a\mathbf{i}' + b\mathbf{j}'$$

between their points. A **transformation** from a linear space $\mathbb{L}$ to a linear space $\mathbb{L}'$ is a function T that pairs each point $\mathbf{P}$ of $\mathbb{L}$ with a point $T(\mathbf{P})$ of $\mathbb{L}'$. A transformation T is a **linear transformation** if

- (i) $\mathbf{P} + \mathbf{Q} = \mathbf{R}$ implies that $T(\mathbf{P}) + T(\mathbf{Q}) = T(\mathbf{R})$ and
- (ii) $a\mathbf{P} = \mathbf{S}$ implies that $aT(\mathbf{P}) = T(\mathbf{S})$.

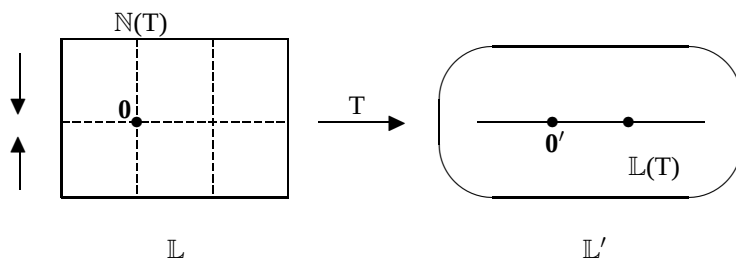


Lemma 74. For a transformation $T : \mathbb{L} \rightarrow \mathbb{L}'$ from $\mathbb{L}$ to $\mathbb{L}'$, the following are equivalent.

- (i) T is a linear transformation.
- (ii) For all $\mathbf{P}, \mathbf{Q}, a$ we have $T(\mathbf{P} + \mathbf{Q}) = T(\mathbf{P}) + T(\mathbf{Q})$ and $T(a\mathbf{P}) = aT(\mathbf{P})$.
- (iii) For all $\mathbf{P}, \mathbf{Q}, a, b$ we have $T(a\mathbf{P} + b\mathbf{Q}) = aT(\mathbf{P}) + bT(\mathbf{Q})$.

[Prove (i) if and only if (ii) if and only if (iii).]

Let $T : \mathbb{L} \rightarrow \mathbb{L}'$ be a linear transformation. The **null space** of T is the subset $\mathcal{N}(T) := \{\mathbf{P} \in \mathbb{L} \mid T(\mathbf{P}) = \mathbf{0}'\}$ and the **range** of T is the subset $\mathcal{R}(T) := \{\mathbf{Q} \in \mathbb{L}' \mid \mathbf{Q} = T(\mathbf{P}) \text{ for some } \mathbf{P} \in \mathbb{L}\}$ of $\mathbb{L}'$.



Problem 75. Define $T : \mathbb{C}[0, 1] \rightarrow \mathbb{R}^3$ by $T(f) = (f(\frac{1}{3}), f(\frac{2}{3}), 0)$. Show that T is a linear transformation and then describe its null space and its range.

Lemma 76. *If $T : \mathbb{L} \rightarrow \mathbb{L}'$ is a linear transformation, then $\mathbb{N}(T)$ is a subspace of $\mathbb{L}$ and $\mathbb{R}(T)$ is a subspace of $\mathbb{L}'$.*

Recall that $T : \mathbb{L} \rightarrow \mathbb{L}'$ is **one-to-one** if for all $\mathbf{P}, \mathbf{Q} \in \mathbb{L}$, we have $\mathbf{P} \neq \mathbf{Q}$ implies $T(\mathbf{P}) \neq T(\mathbf{Q})$, and T is **onto** $\mathbb{L}'$ if for all $\mathbf{P}' \in \mathbb{L}'$ there is a $\mathbf{P} \in \mathbb{L}$ such that $T(\mathbf{P}) = \mathbf{P}'$.

Lemma 77. *A linear transformation $T : \mathbb{L} \rightarrow \mathbb{L}'$ is one-to-one if and only if $\mathbb{N}(T) = \{\mathbf{0}\}$, and T is onto if and only if $\mathbb{R}(T) = \mathbb{L}'$.*

A linear transformation $T : \mathbb{L} \rightarrow \mathbb{L}'$ is called a **linear space isomorphism** if it is one-to-one and onto. A linear space isomorphism $T : \mathbb{L} \rightarrow \mathbb{L}'$ is an **inner product space isomorphism** if it has the additional property that

$$(iii) \quad \mathbf{P} \cdot \mathbf{Q} = T(\mathbf{P}) \cdot T(\mathbf{Q}) \text{ for all } \mathbf{P}, \mathbf{Q} \in \mathbb{L}.$$

We say that $\mathbb{L}$ and $\mathbb{L}'$ are **isomorphic** if there is an isomorphism $T : \mathbb{L} \rightarrow \mathbb{L}'$ from $\mathbb{L}$ onto $\mathbb{L}'$. What we saw to be true of two dimensional inner product spaces extends to n -dimensional inner product spaces.

Representation Theorem 78. *If $\mathbb{L}$ is an n -dimensional linear (inner product) space, then there is an (inner product space) isomorphism $T : \mathbb{R}^n \rightarrow \mathbb{L}$.*

In this sense, $\mathbb{R}^n$ is the only n -dimensional (inner product) space.

In the remainder of this chapter we consider only linear spaces, without reference to inner product. Our next theorem, which applies to all finite dimensional linear spaces, draws on a broad range of the ideas that we have discussed in this course.

Dimension Theorem 79. *Let $\mathbb{L}$ and $\mathbb{L}'$ be finite dimensional linear spaces and assume that $T : \mathbb{L} \rightarrow \mathbb{L}'$ is a linear transformation. Then*

$$\text{dimension}(\mathbb{L}) = \text{dimension}(\mathbb{N}(T)) + \text{dimension}(\mathbb{R}(T)).$$

[Extend a basis for $\mathbb{N}(T)$ to a basis for $\mathbb{L}$ and use this extended basis to find a basis for $\mathbb{R}(T)$.]

Let $T : \mathbb{L} \rightarrow \mathbb{L}'$ be a linear transformation between linear spaces $\mathbb{L}$ and $\mathbb{L}'$ with bases $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_m\}$ and $\mathbb{B}' = \{\mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_n\}$, respectively. If

$$\mathbf{P} = c_1\mathbf{P}_1 + c_2\mathbf{P}_2 + \dots + c_m\mathbf{P}_m$$

is a point of $\mathbb{L}$, we have

$$T(\mathbf{P}) = c_1T(\mathbf{P}_1) + c_2T(\mathbf{P}_2) + \dots + c_mT(\mathbf{P}_m).$$

Thus the linear transformation T is completely determined by its values on the m basis points of $\mathbb{L}$. For $j = 1, \dots, m$, let

$$T(\mathbf{P}_j) = a_{1j}\mathbf{Q}_1 + a_{2j}\mathbf{Q}_2 + \dots + a_{nj}\mathbf{Q}_n.$$

Then T is completely determined by the values of the nm numbers in the $n \times m$ **matrix**

$$M_T := \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}$$

It turns out to be convenient to use the $m \times 1$ column matrix

$$\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}$$

to denote the point $\mathbf{P} = c_1\mathbf{P}_1 + c_2\mathbf{P}_2 + \cdots + c_m\mathbf{P}_m$ of $\mathbb{L}$ and, similarly, to use a $n \times 1$ column matrix to denote a point of $\mathbb{L}'$. In this notation the linear transformation T is given simply by matrix multiplication.

Matrix Theorem 80. *Let $T : \mathbb{L} \rightarrow \mathbb{L}'$ be a linear transformation from an m -dimensional linear space $\mathbb{L}$ to an n -dimensional linear space $\mathbb{L}'$ with bases $\mathbb{B}$ and $\mathbb{B}'$ as above. Then T is given by matrix multiplication as*

$$T \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} c_1a_{11} + c_2a_{12} + \cdots + c_ma_{1m} \\ c_1a_{21} + c_2a_{22} + \cdots + c_ma_{2m} \\ \vdots \\ c_1a_{n1} + c_2a_{n2} + \cdots + c_ma_{nm} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}$$

or, stated more concisely, $T(\mathbf{P}) = M_T\mathbf{P}$. Conversely, if M is any $n \times m$ matrix, then the matrix equation $T(\mathbf{P}) := M\mathbf{P}$ defines a linear transformation from $\mathbb{L}$ to $\mathbb{L}'$.

In this way linear transformations from $\mathbb{L}$ to $\mathbb{L}'$ correspond exactly to $n \times m$ matrices.

Mathematicians have a peculiar way to multiply matrices. The **product** of a $p \times n$ matrix and an $n \times m$ matrix is the $p \times m$ matrix given by

$$\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ b_{i1} & b_{i2} & \cdots & b_{in} \\ \vdots & \vdots & \cdots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{bmatrix} \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1m} \\ a_{21} & \cdots & a_{2j} & \cdots & a_{2m} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nm} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1m} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{im} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ c_{p1} & \cdots & c_{pj} & \cdots & c_{pm} \end{bmatrix}$$

where

$$c_{ij} = b_{i1}a_{1j} + b_{i2}a_{2j} + \cdots + b_{in}a_{nj}.$$

In spite of this strange way of multiplying matrices, it turns out that matrix multiplication is in general associative. Prove this in the 2×2 case, which is already a bit messy.

Lemma 81. *Multiplication of 2×2 matrices is associative.*

Consider three finite dimensional linear spaces, $\mathbb{L}$, $\mathbb{L}'$ and $\mathbb{L}''$. Assume they have bases $\mathbb{B} = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_m\}$, $\mathbb{B}' = \{\mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_n\}$, $\mathbb{B}'' = \{\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_p\}$, respectively. Let

$$T : \mathbb{L} \rightarrow \mathbb{L}' \quad \text{and} \quad U : \mathbb{L}' \rightarrow \mathbb{L}''$$

be linear transformations with matrices

$$M_T = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1m} \\ a_{21} & \cdots & a_{2j} & \cdots & a_{2m} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nm} \end{bmatrix} \quad \text{and} \quad M_U = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ b_{i1} & b_{i2} & \cdots & b_{in} \\ \vdots & \vdots & \cdots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{bmatrix}.$$

What is the matrix for the composition $U \circ T : \mathbb{L} \rightarrow \mathbb{L}''$?

Composition Theorem 82. $M_{U \circ T} = M_U M_T$, that is, multiplication of matrices corresponds exactly to composition of the corresponding linear transformations.

This theorem answers two questions. It explains why we choose this complicated way of multiplying matrices. It also explains why this complicated multiplication should turn out to be associative. It is because composition of functions is *always* associative:

$$\begin{aligned} ((f \circ g) \circ h)(x) &= (f \circ g)(h(x)) = f(g(h(x))) \\ (f \circ (g \circ h))(x) &= f((g \circ h)(x)) = f(g(h(x))); \end{aligned}$$

Both simply ask us to apply h , then g , then f !